Frequency Tables and Graphs by Data Type

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Descriptive statistics.
- **Compare with:** Cumulative distribution function (CDF).

1. Overview

Key point: A categorical feature is summarised by a frequency table and its relative and cumulative versions; a numerical feature by a histogram; two or more features by graphs chosen from their types.

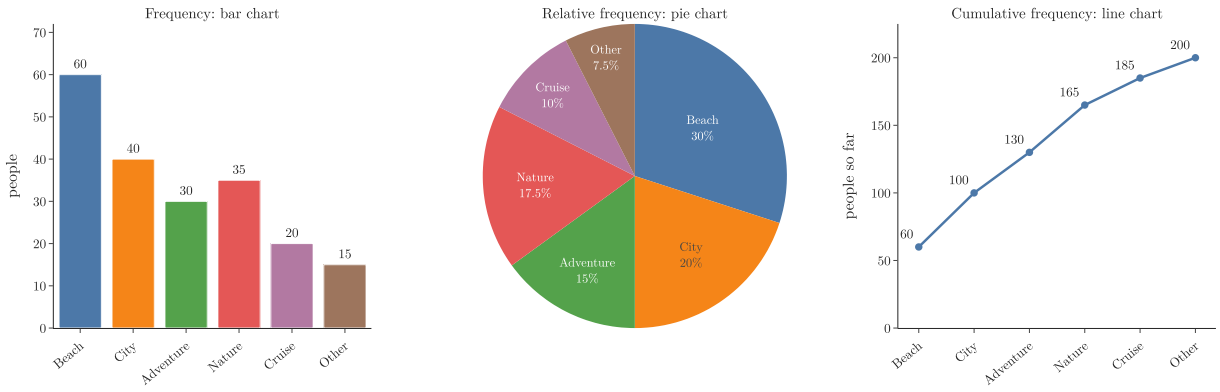


Figure 1: A frequency table drawn three ways: frequency as a bar chart, relative frequency as a pie chart, cumulative frequency as a line chart

Summarising data and presenting it in a meaningful way is **descriptive statistics** (G-596). Figure 1 shows the three tables we can build from one categorical feature, each with the chart that suits it. A **feature** (G-772) is one variable of the data, one column of the table; an **observation** (G-1374) is one record, one row of the table.

This Note covers:

- the three frequency tables of one categorical feature, and the **dot plot** (G-2230) of a discrete one (section 2);
- the histogram of one numerical feature, and the shapes it can take (section 3);
- the graphs for two features (section 4);
- the graphs for more than two features (section 5).

2. Frequency tables for a categorical feature

> **Key point:** Count each category (frequency), turn the counts into shares (**relative frequency** (G-1664)), and add them up as we go (cumulative frequency).

2.1 Frequency distribution table

> **Key point:** A frequency distribution table lists each category with the number of times it occurs.

We survey 200 people about their favourite type of vacation. Each answer is one of six categories: beach, city, adventure, nature, cruise or other. The data has one **observation** (G-1374) per person, with their name and their answer.

A **frequency distribution table** (G-807) summarises how many times each value occurs in a dataset. For the survey, we count each category:

Vacation	Frequency	Relative frequency	Cumulative frequency
Beach	60	0.300	60
City	40	0.200	100
Adventure	30	0.150	130
Nature	35	0.175	165
Cruise	20	0.100	185
Other	15	0.075	200
Total	**200**	**1.000**	

Figure 2 builds the table one category at a time. Watch the left panel fill in the counts, and the right panel keep a running total that ends at 200.

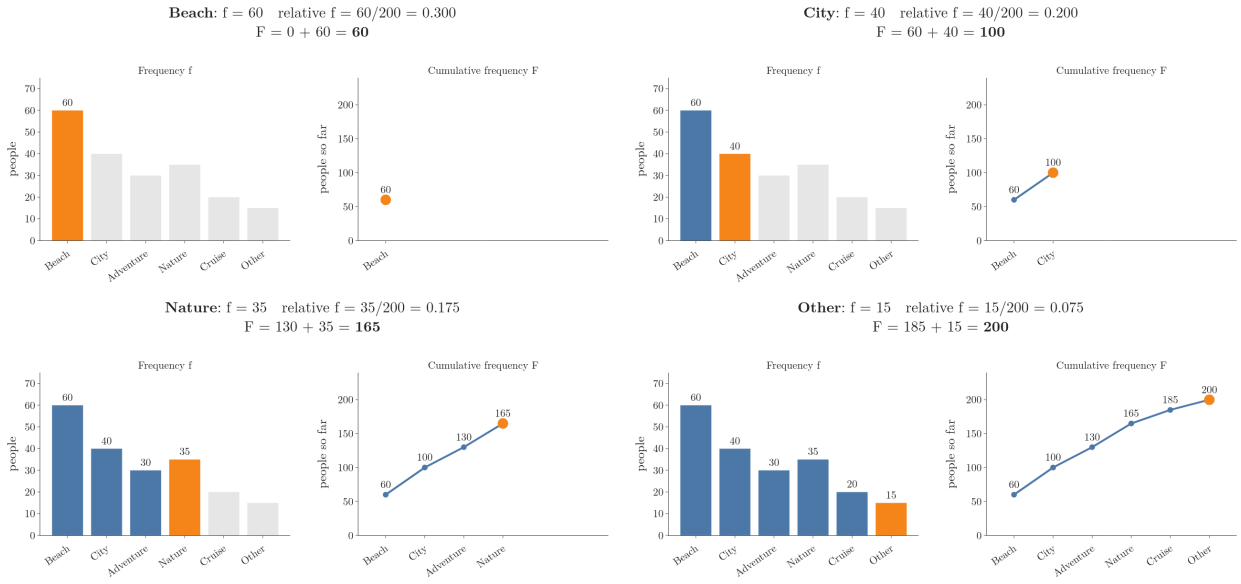


Figure 2: The vacation table built step by step. Each step counts one category (orange bar), divides it by 200 for the relative frequency, and adds it to the running total on the right, which ends at 200.

The frequencies are drawn as a bar chart, one bar per category (Figure 1, left). This bar chart is the count plot, and pandas (the Python table library) builds the table with `value_counts()`.

2.2 Relative frequency

> **Key point:** Relative frequency is a category's share of all the data: its frequency divided by the total.

The **relative frequency** (G-1664) of a category is its proportion, or percentage, of the data.

1. **In words:** divide the category's count by the total number of **observations** (G-1374).
2. **Formula:** for a category with frequency f out of n observations,

$$\text{relative frequency} = \frac{f}{n}$$

3. **Example:** 60 of the 200 people chose the beach, so

$$\text{relative frequency of Beach} = \frac{60}{200}$$

$$\text{relative frequency of Beach} = 0.30$$

That is 30 percent.

The relative frequencies always add up to 1 (100%). Summing to 1 makes them the natural input for a **pie chart** (G-1495), whose slices show each category's share (Figure 1, middle).

2.3 Cumulative frequency

> **Key point:** Cumulative frequency is the running total of the frequencies, so the last value is always the total.

The **cumulative frequency** (G-517) of a category is its frequency plus the frequencies of all the categories before it.

1. **In words:** walk down the table, keeping a running total of the frequencies.
2. **Formula:** for categories in the table's order with frequencies $f_1, f_2, \ldots$, the cumulative frequency of the k-th category is

$$F_k = f_1 + f_2 + \cdots + f_k$$

3. **Example:** walk down the vacation table, one category per line:

$$F_{\text{Beach}} = 60$$

$$F_{\text{City}} = 60 + 40 = 100$$

$$F_{\text{Adventure}} = 100 + 30 = 130$$

$$F_{\text{Nature}} = 130 + 35 = 165$$

$$F_{\text{Cruise}} = 165 + 20 = 185$$

$$F_{\text{Other}} = 185 + 15 = 200$$

   The last value is the total, 200.

A line chart shows the running total (Figure 1, right). The same running total of the relative frequencies, the **cumulative relative frequency** (G-518), climbs from 0.30 to 0.50, and so on, to 1.

Cumulative frequency is most meaningful when the categories have an order, such as ordinal data or the **bins** (G-298) of a histogram: "how many people scored 60 or less?". For unordered categories it depends on the order we list them in.

Python: The three tables with pandas.

```
freq = df["vacation"].value_counts(sort=False)
rel = freq / freq.sum()          # relative frequency
cum = freq.cumsum()              # cumulative frequency
```

`value_counts(normalize=True)` gives the relative frequencies directly. `cumsum()` is the running total. The Notebook (`MA-007-frequency-tables-and-graphs.ipynb`) builds all three tables and charts.

2.4 A discrete feature: the dot plot

Key point: A numerical feature with only a few separate values gets the same frequency table; drawn with one dot per **observation** (G-1374), the table is a dot plot.

A discrete **feature** (G-772), such as the number of people in a restaurant party, has few distinct values, so each value can play the part of a category. For the 76 Sunday parties of the tips data (a standard dataset of 244 restaurant bills with the tip, party size and day) we count each party size:

Party size	1	2	3	4	5	6
Frequency	0	39	15	18	3	1

A **dot plot** (G-2230) draws this table with one dot per observation, stacked above its value (Figure 3). The table, the dot plot and the raw list of 76 numbers all hold the same data; the dot plot is the quickest to read.

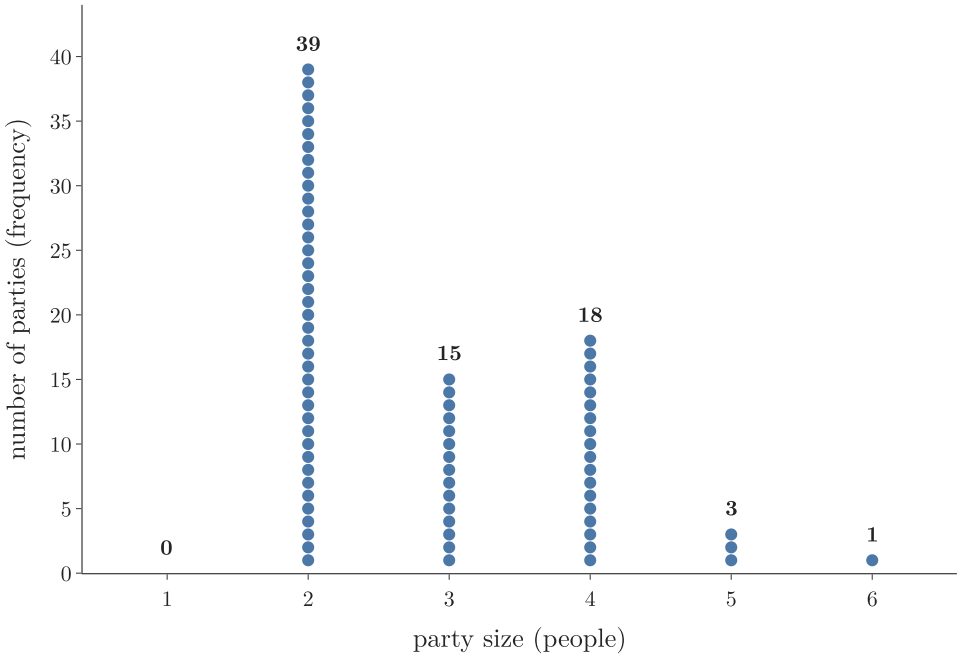


Figure 3: Dot plot of party size for the 76 Sunday parties: one dot per party, stacked over its size. Idea after Khan Academy, "Frequency tables and dot plots"

Three questions answered straight from Figure 3:

- **Which size is most frequent?** The tallest stack: 2 people, with 39 parties. That value is the **mode** (the most frequent value).
- **What is the range?** The largest size minus the smallest:

$$6 - 2 = 4$$

- **How many parties had more than 4 people?** Add the stacks to the right of 4:

$$3 + 1 = 4$$

  That is 4 parties.

3. Histograms for a numerical feature

> **Key point:** A numerical **feature** (G-772) has no categories, so we make our own by cutting its range into bins; the histogram is the frequency table of those bins.

A **dot plot** (G-2230) fails for a measured feature. Figure 4 starts with the 244 total bills of the tips data, one dot per bill on a number line:

1. **The dots hide each other.** Many bills are close together, so the dots overlap and we cannot see how many there are.
2. **Stacking equal values does not help.** A bill of 13.42 dollars rarely occurs twice. Only 15 of the 244 dots have an equal partner to stack on.
3. **So we stack by range instead.** We cut the line into intervals 5 dollars wide and stack every dot that falls in the same interval. The stacks are the bars of a histogram.

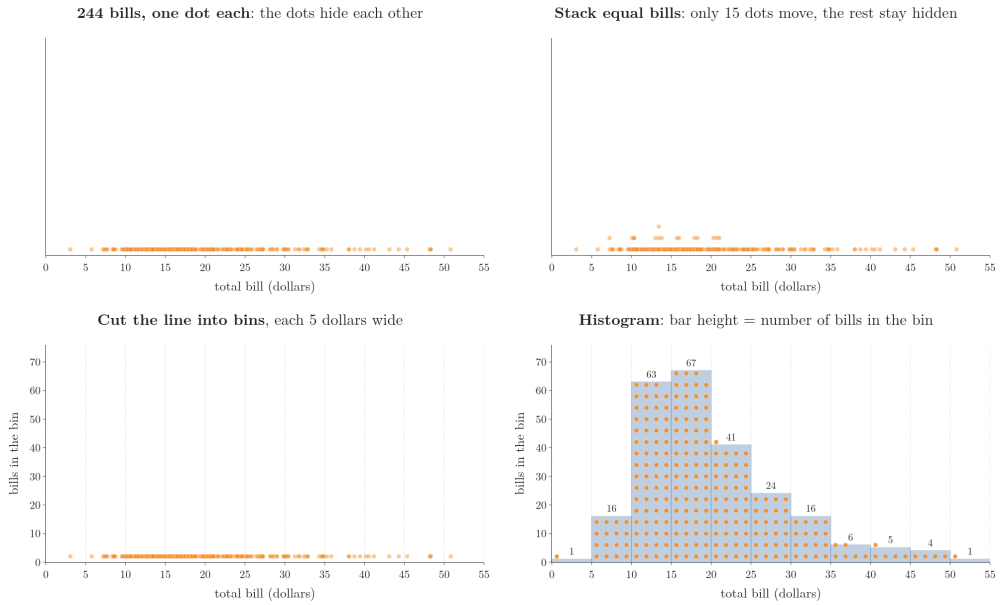


Figure 4: The 244 total bills as dots on a line, then stacked by equal value, then cut into bins 5 dollars wide and stacked per bin: the stacks are the histogram. Idea after StatQuest, "Histograms, Clearly Explained"

The tallest stack in Figure 4 holds the 67 bills between 15 and 20 dollars; only 1 bill is above 50. So a histogram also says where the next **observation** (G-1374) will probably fall: in a tall bin, rarely out in the short ones.

In general, a numerical feature such as a bill or an age has no categories to count. So we create them, in three steps:

1. Cut the range into equal intervals, such as 0 to 10, 10 to 20, 20 to 30. Each interval is a **bin** (G-298), also called a bucket.
2. Put each person in the bin that holds their age.
3. Count the people in each bin.

The result is a **frequency distribution table** (G-807) on bins instead of categories, and its chart is the **histogram** (G-899; see the histogram).

A histogram looks like a bar chart with one difference: its bars touch. The categories of a bar chart are separate; the bins of a histogram are continuous ranges, one ending where the next begins.

The number of bins changes the picture, as choosing the number of bins shows. Figure 5 cuts the 714 known Titanic ages into 2, 8, 20 and 80 bins:

- **2 bins:** each bar covers 40 years of ages, so the shape is hidden.
- **8 and 20 bins:** the peak in the twenties and the long tail of older passengers appear.
- **80 bins:** each bar holds only a handful of people, so the bars jump up and down by chance.

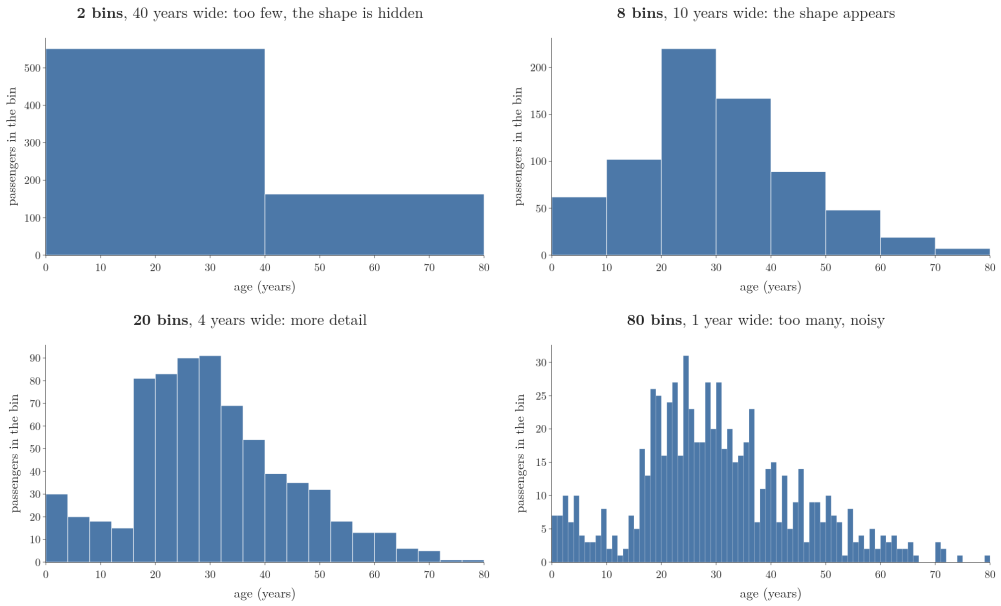


Figure 5: The same 714 Titanic ages in 2, 8, 20 and 80 bins. The data never changes; only the bin width does.

We try a few bin counts and keep the clearest; the default of a plotting library is only a starting point.

A histogram trades detail for shape. Figure 4 shows that 67 bills lie between 15 and 20 dollars, but once the dots are replaced by a bar, the exact bills are gone. So a histogram answers "how many values are above 30?" by adding bars, yet it cannot give an exact median (the middle value of the sorted list), which needs the sorted values themselves. A dot plot keeps every value, and the box plot marks the median directly.

3.1 The shapes of a histogram

> **Key point:** Histograms come in a few typical shapes: symmetric, bimodal, skewed to the right or left, uniform, or no clear pattern.

Figure 6 shows six typical shapes.

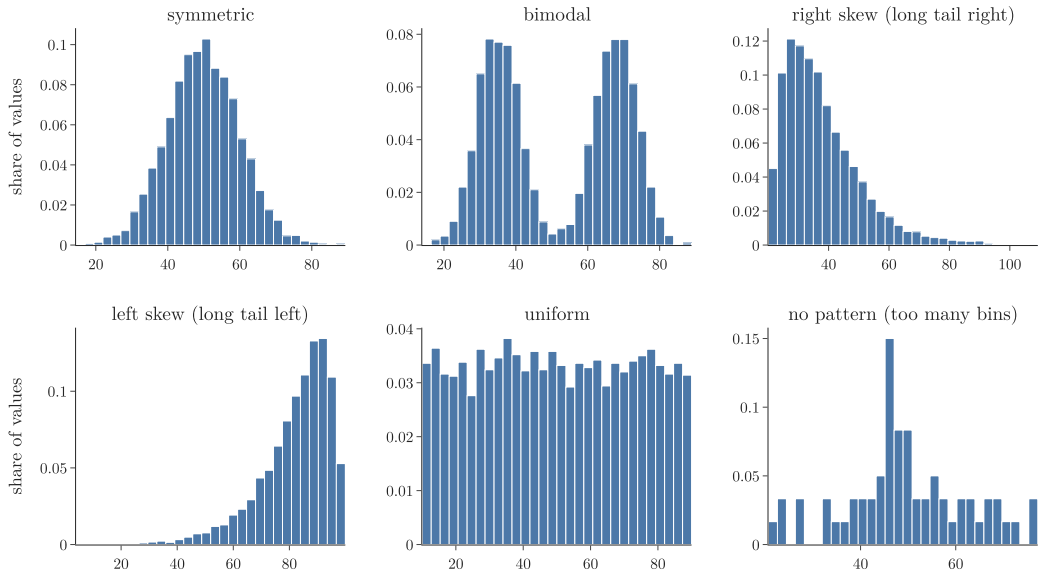


Figure 6: Six shapes a histogram can take

- **Symmetric:** most values in the middle, fewer and fewer towards both sides.
- **Bimodal** (G-296): two separate peaks, two groups of values where points are dense. With three peaks it is trimodal. Two peaks often mean two groups mixed together, each with its own centre, such as the heights of children and adults (NIST Handbook §1.3.3.14.5).
- **Right or left skew:** a long tail on one side; the shapes and the skewness number (how lopsided a distribution is) are taught in **skewness** (G-1817).
- **Uniform:** every **bin** (G-298) holds about the same number of values. Too few bins also make data look uniform.
- **No pattern:** the bars jump up and down. Usually there are too many bins for the amount of data, here 30 bins for 60 values.

Named shapes such as the normal and uniform distributions come in the probability distribution Notes.

> **Extra:** Remember the direction of skew by the tail, not by the hump. In a right-skewed histogram the hump is on the left and the long tail points right.

4. Graphs for two features

> **Key point:** Two features are categorical + categorical (contingency table), numerical + numerical (scatter plot) or categorical + numerical (an **aggregate** (G-182) per category, or a contingency table of bins).

Studying two **features** (G-772) together is **bivariate analysis** (G-310); four datasets for four kinds of plots maps each pair of feature types to its plots.

Figure 7 draws one example of each pair on the Titanic.

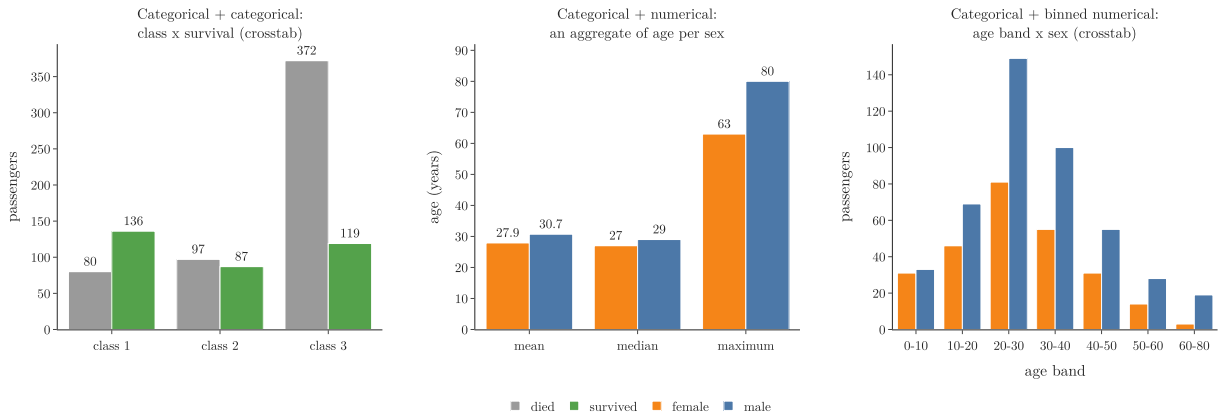


Figure 7: Left: a contingency table of class against survival, drawn as grouped bars. Middle: three aggregates of age for each sex. Right: a contingency table of age band against sex.

- **Categorical + categorical:** a **contingency table** (G-464), also called a **crosstab** (G-511): the counts for every pair of categories (see crosstab and heatmap). On the Titanic, a contingency table of class against survival shows 372 of the 549 passengers who died were in third class (Figure 7, left). A side-by-side or stacked bar chart draws it.
- **Numerical + numerical:** a scatter plot (one dot per **observation** (G-1374), with one feature on each axis; see total bill against tip). Its pattern can show a positive relation (one rises as the other rises), a negative one, or none, which **correlation** (G-490) measures.
- **Categorical + numerical:** two options, below.

4.1 Categorical and numerical: any aggregate per category

> **Key point:** A bar per category whose height is any summary of the numerical **feature** (G-772): mean, median, maximum, standard deviation.

A bar chart of a categorical feature against a numerical one does not show counts. Each bar shows an **aggregate** (G-182), one summary number computed from the numerical values in that category. The bar plot uses the mean, but any summary works:

Sex	Mean age	Median age	Maximum age
Female	27.9	27	63
Male	30.7	29	80

The mean bars compare typical ages; the maximum bars show that the oldest man (80) was much older than the oldest woman (63) (Figure 7, middle).

4.2 Categorical and numerical: a contingency table of bins

> **Key point:** Cutting the numerical feature into **bins** (G-298) turns it into a categorical one, so a contingency table works.

We can also bin the numerical **feature** (G-772), as for a histogram, and then count every pair of age band and category:

Age band	Female	Male
0 to 10	31	33
10 to 20	46	69
20 to 30	81	149
30 to 40	55	100
40 to 50	31	55
50 to 60	14	28
60 to 80	3	19

The cell "20 to 30, male" means 149 passengers were men aged over 20 and up to 30. The table shows the men outnumbered the women in every band, most of all among the older passengers (19 against 3 over 60). Figure 7, right, draws the table: the blue bars are taller in every band.

> **Python:** Binning, then a **contingency table** (G-464).
> ```
> bands = pd.cut(df["Age"], bins=[0, 10, 20, 30, 40, 50, 60, 80])
> pd.crosstab(bands, df["Sex"])
> ```
> `pd.cut` puts each age in its band; a band such as (20, 30] includes 30 but not 20.

> **Extra:** A **crosstab** (G-511) only counts pairs of categories. A **pivot table** (G-1500) can also **aggregate** (G-182) a third, numerical feature for each pair: for example, the mean age of the male passengers in first class. Spreadsheets and pandas (`pivot_table`) both offer it; a pivot table and heatmap uses one.

5. Graphs for more than two features

> **Key point:** A **3D scatter plot** (G-50), colour (hue), facet grids, pair plots and bubble charts each fit three or more features into one figure.

Studying more than two **features** (G-772) at once is **multivariate analysis** (G-1280). Most of its graphs are bivariate graphs with an extra feature added:

- **Hue** (G-906): colour shows a categorical feature, on a scatter plot, bar plot or box plot. A bar plot of mean age by sex with hue = class shows three features. See adding more columns: colour, marker and size.
- **Bubble chart:** a scatter plot whose dot size shows a third numerical feature, such as countries by GDP and population with bubble size for literacy rate (the "size" setting

of the same section above).

- **Pair plot:** a scatter plot for every pair of numerical features, with a histogram on the diagonal (see the pair plot).
- **3D scatter plot** and **facet grid:** below.

5.1 The 3D scatter plot

> **Key point:** Three numerical **features** (G-772) on three axes; colour can add a fourth.

A **3D scatter plot** (G-50) places each **observation** (G-1374) as a point in three dimensions, one numerical feature per axis. Figure 8 plots the tips data: total bill, party size and tip, with lunch and dinner by colour.

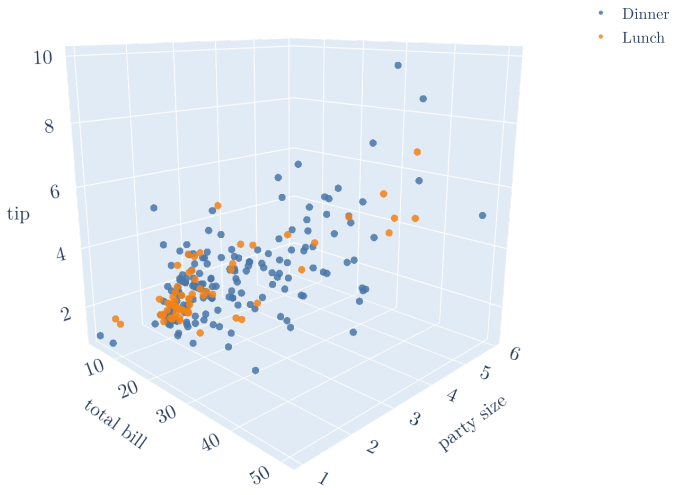


Figure 8: 3D scatter plot of total bill, party size and tip

Bigger bills and bigger parties go with bigger tips: the points rise towards the far corner. On paper a 3D plot is hard to read; in the Notebook it can be rotated with the mouse, which is where it is useful.

5.2 The facet grid

> **Key point:** The same plot repeated side by side, one panel per category of another **feature** (G-772).

A **facet grid** (G-744) splits the data by a categorical feature and draws the same plot for each part, side by side, on shared axes. Figure 9 draws bill against tip once for lunch and once for dinner, with smokers in orange.

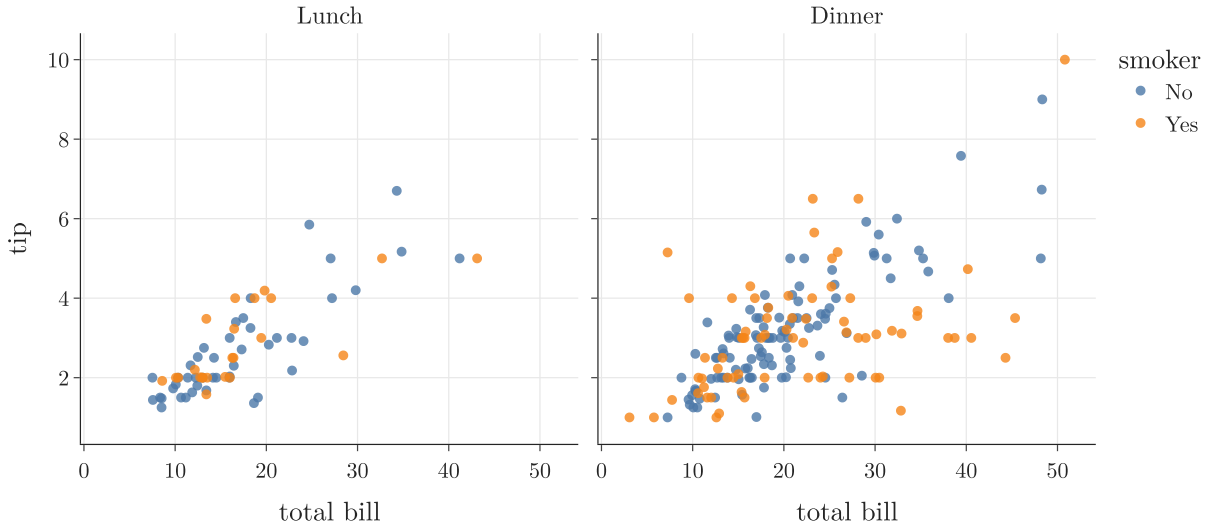


Figure 9: Bill against tip, one panel per meal time, smokers by colour

Figure 9 shows four features at once: bill, tip, meal time (the panels) and smoker (the colour). Dinner has more customers and more of the large bills.

Python: 3D scatter plot (G-50) and facet grid.

```
import plotly.express as px

px.scatter_3d(tips, x="total_bill", y="size", z="tip", color="time")
px.scatter(tips, x="total_bill", y="tip", color="smoker",
           facet_col="time")
```

In seaborn, `sns.relplot(data=tips, x="total_bill", y="tip", hue="smoker", col="time")` draws the facet grid.

6. Summary

Features	Types	Table	Graph	Why it matters
One	categorical	frequency, relative, cumulative frequency	bar chart, pie chart, line chart	answers how many, what share, and how many so far
One	numerical	frequency table of bins	histogram	a measured feature has no categories, so bins make them
Two	categorical + categorical	contingency table (crosstab)	side-by-side or stacked bars	counts every pair, such as class against survival
Two	numerical + numerical	none	scatter plot	shows a positive, negative or no relation
Two	categorical + numerical	aggregate per category, or crosstab of bins	bar chart of an aggregate, box plots	compares a summary of the numbers across categories
Three or more	any	pivot table	hue, bubble chart, pair plot, 3D scatter, facet grid	colour, size or panels fit the extra features into one figure

- Relative frequency = frequency / total; cumulative frequency = running total. The shares add up to 1, so they suit a **pie chart** (G-1495); the running total answers questions such as "how many scored 60 or less?".
- Histogram bars touch because **bins** (G-298) are continuous ranges.
- Histogram shapes: symmetric, **bimodal** (G-296), right skew, left skew, uniform, no pattern. Two peaks often mean two groups mixed together, and too few or too many bins can fake a uniform or patternless shape, so try a few bin counts.
- The types of the **features** (G-772) decide the table and the graph, so check the types first.

7. Sources

Built from

- CampusX, "Session 38 - Descriptive Statistics Part 1 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=Uv3Blie7F3g
- StatQuest with Josh Starmer, "StatQuest: Histograms, Clearly Explained", YouTube, https://www.youtube.com/watch?v=qBigTkBLU6g

- Khan Academy, "Frequency tables and dot plots", YouTube, https://www.youtube.com/watch?v=gdE4
- Khan Academy, "Comparing dot plots, histograms, and box plots", YouTube, https://www.youtube.com/watch?v=s_w3EJ2Jzw0

Other references

- NIST/SEMATECH. *e-Handbook of Statistical Methods*, section 1.3.3.14.5, Histogram Interpretation: Bimodal Mixture of 2 Normals. itl.nist.gov, div898 handbook, histogr5.

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Dot plot (G-2230)	A frequency table drawn with one dot per observation, stacked over its value.
Relative frequency (G-1664)	A category's share of all the data: its frequency divided by the total.
Frequency distribution table (G-807)	A table of each value or category with the number of times it occurs.
Cumulative frequency (G-517)	The running total of the frequencies, up to and including a category.
Cumulative relative frequency (G-518)	The running total of the relative frequencies; ends at 1.
Aggregate (G-182)	One summary number (mean, median, maximum, ...) computed from a group of values.
3D scatter plot (G-50)	A scatter plot with three numerical features on three axes, one point per observation, to see how the three relate at once.
Hue, style, size (G-906)	Plot settings that show an extra column by colour, marker shape or dot size.
Facet grid (G-744)	The same plot repeated side by side, one panel per category of another column.
Descriptive statistics (G-596)	Numbers that summarise data, such as count, mean, spread and percentiles.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Pie chart (G-1495)	A circle split into slices sized by each category's share.
Bin (G-298)	One of the equal ranges a histogram splits the data into.

Term	Meaning
Histogram (G-899)	A bar chart of how many values fall in each equal range (bin) of a numerical column; it shows the shape of a distribution at a glance: where values pile up, how far they spread and whether there are several peaks.
Bimodal (G-296)	A distribution with two peaks.
Skewness (G-1817)	A number for how lopsided a distribution is: 0 symmetric, positive right tail, negative left tail.
Bivariate analysis (G-310)	Studying two columns together to see how they are related, for example with a scatter plot or a bar plot.
Contingency table (G-464)	A table that counts the observations for every pair of categories of two categorical features, so we can see whether the two are related; also called a crosstab.
Crosstab (G-511)	A table that counts the rows for every pair of categories of two categorical columns, so we can see whether the two are related; pandas builds it with `pd.crosstab`. Also called a contingency table.
Correlation (G-490)	A number from -1 to +1 that says how strongly two numerical features rise or fall together along a straight line: the covariance divided by both standard deviations, so the units drop out.
Pivot table (G-1500)	A grid with one column's values as rows, another's as columns, and a third in the cells.
Multivariate analysis (G-1280)	Studying more than two variables together.